

SIH26034 - Legal Metrology Automated Compliance System

Complete UI/UX Specification, Design Tokens, Screen Wireframes & Interaction Workflows

1. Document Control & Design Philosophy

- **Standard:** SIH26034 Production-Grade Design Specification
 - **Design Directive:** "Evidence-Grade Ergonomics, High-Cognitive-Clarity, Zero-Ambiguity Adjudication"
 - **Target Users:**
 1. *Field Legal Metrology Inspector (LMO/LMI):* Rugged mobile/tablet usage in noisy markets, grocery godowns, retail stores, and packaging factories. Needs high-contrast, large touch targets (≥ 48 dp), one-handed capture controls, and real-time capture guidance.
 2. *Senior Adjudicating Officer / District Controller:* High-density dual-monitor desktop workstation. Needs side-by-side high-resolution optical zoom, multi-layered visual bounding overlays, OCR text inspection, and instantaneous override/audit logging.
 3. *State Metrology Directorate:* High-level oversight dashboard displaying state-wide compliance statistics, recurring offender indices, seizure trends, and e-commerce platform compliance rates.
 4. *Packer / Importer / Manufacturer:* Self-audit portal for pre-market label validation, report download, and notice rectification submissions.
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2. Design System & Foundational Design Tokens

Typography:

Display / Headers: "Inter", -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, sans-serif

Monospace / Data: "JetBrains Mono", "Fira Code", monospace (For timestamps, hashes, OCR strings, font heights)

Base Scale:

- Display 1: 32px / Line Height: 40px (Bold 700)
- Heading 1: 24px / Line Height: 32px (Semi-Bold 600)
- Heading 2: 20px / Line Height: 28px (Semi-Bold 600)
- Subheading: 16px / Line Height: 24px (Medium 500)
- Body Regular: 14px / Line Height: 20px (Regular 400)
- Body Small / Metadata: 12px / Line Height: 16px (Regular 400)
- Legal Monospace: 12px / Line Height: 18px (Medium 500)

Color Tokens:

Brand / Institutional:

- Primary Blue (Govt / Official): #1B365D (Deep Ashoka Navy)
- Primary Blue Accent: #2E5B9A
- Surface Background Light: #F8FAFC (Slate 50)
- Card / Panel Surface Light: #FFFFFF
- Surface Background Dark: #0F172A (Slate 900)
- Card / Panel Surface Dark: #1E293B (Slate 800)

Compliance Semantic Status Colors:

- Fully Compliant (PASS): #059669 (Emerald 600) / Surface Tint: #ECFDF5
- Partial / Minor Infraction (WARNING): #D97706 (Amber 600) / Surface Tint: #FFFBEF

Non-Compliant / Violation (FAIL): #DC2626 (Rose 600) / Surface Tint: #FEF2F2

- Pending Inspection / Review (NEUTRAL): #475569 (Slate 600) / Surface Tint: #F1F5F9

High-Confidence Bounding Box: #10B981 (Green)

- Medium-Confidence Bounding Box: #F59E0B (Amber)
- Low-Confidence / Flagged Bounding Box: #EF4444 (Crimson)
- Reference Calibration Ruler / ArUco Box: #6366F1 (Indigo 500)

Elevation & Spacing:

Grid System: 8-point base grid (4px, 8px, 16px, 24px, 32px, 48px, 64px)

Border Radii: Sharp-Clean (Inputs: 4px, Cards: 8px, Modals: 12px, Badges: 9999px)

Elevation Shadows:

- Level 0: None
- Level 1 (Card Rest): 0 1px 3px rgba(0, 0, 0, 0.08), 0 1px 2px rgba(0, 0, 0, 0.04)
- Level 2 (Card Hover / Active): 0 4px 6px -1px rgba(0, 0, 0, 0.1), 0 2px 4px -1px rgba(0, 0, 0, 0.06)
- Level 3 (Modal / Floating Inspector HUD): 0 10px 15px -3px rgba(0, 0, 0, 0.15)

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|                                     |
|                               SCREEN NAVIGATION FLOW                       |
|                                     |
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|                                     |
| [0. Auth / Role Gate]             |
|                                     |
|   |                                 | |
|   |→ [1. Executive Dashboard] → [6. Batch Ingestion / Analytics]          |
|   |                                 |
|   |→ [2. Inspector Mobile Capture HUD] ─┐                                |
|   |                                     | (Scan / Ingest)                  |
|   |→ [4. E-Commerce Listing Auditor] —┬─> [3. Verification Canvas]         |
|   |                                     |                                   |
|   |                                     ▼ (Adjudicate)                      |
|   |                                     [5. Legal Notice Generator]           |
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4.1 Screen 1: Executive Dashboard (/dashboard)

- **Primary Goal:** Real-time visibility into inspections, compliance heatmaps, and enforcement pipeline.

- Ministry Seal & "Legal Metrology Enforcement Directorate"
- Current Active Inspector / Officer Badge, District Name, Active Jurisdiction.
- Quick Search (Barcodes, Notice IDs, Brand Names, E-commerce URL).
- "New Field Scan" & "Ingest E-Commerce URL" Action CTA Buttons.

1. *Total Inspections Logged (30 Days)*: Total count + week-on-week trend.
2. *Compliance Rate*: Percentage (e.g. 78.4%) with green/amber/red status gauge.
3. *Violations Detected*: Categorized by Font Height, Missing MRP, Unit Sale Price, Origin.
4. *Active Legal Notices Issued*: Drafts, Pending Signatures, Dispatched under Section 36(1).

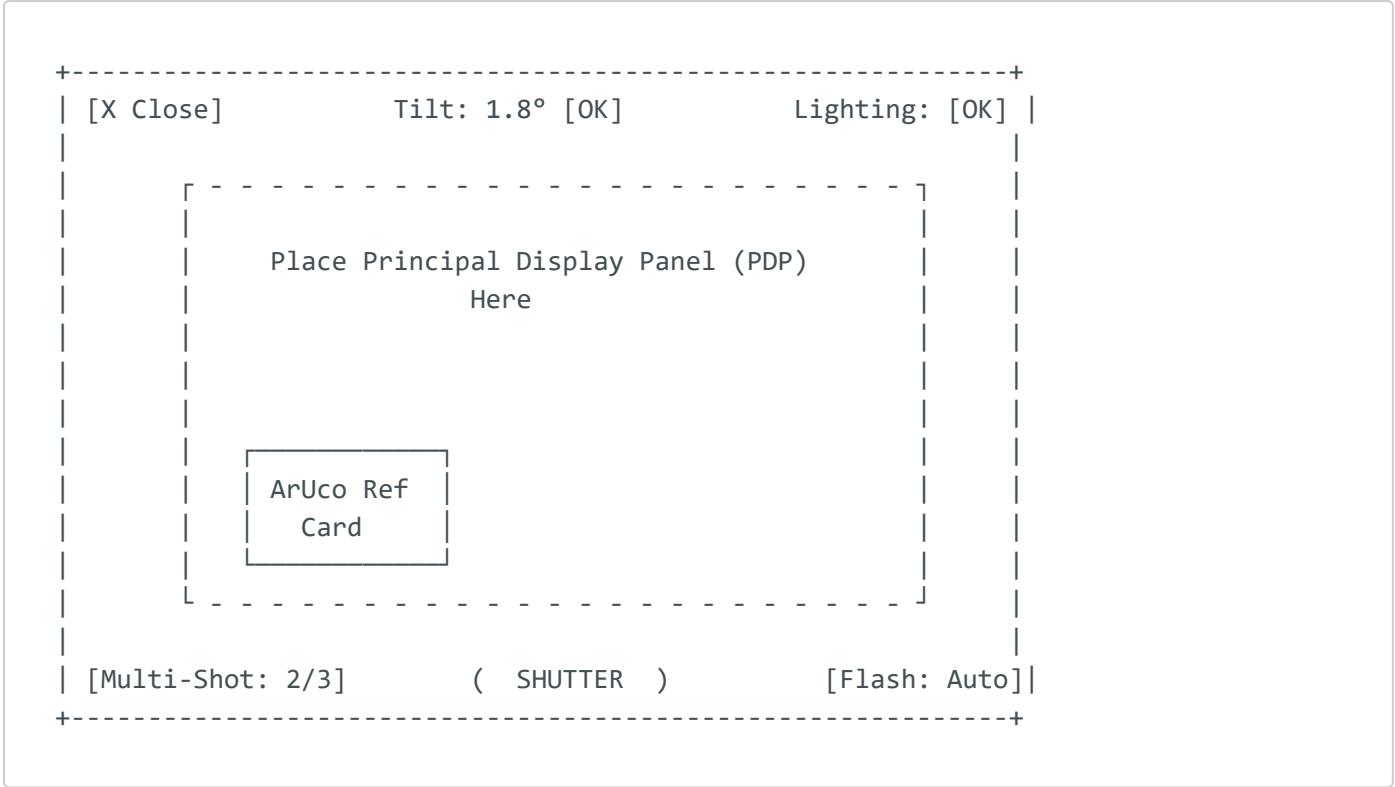
- **Main Viewport Layout:**

- *Left 65%*: Recent Inspections Table. Columns: Inspection ID, Timestamp, Brand/Product, Category, Capture Mode (Field / E-Comm), Primary Violation, Overall Status (Badge: PASS / FAIL / FLAG), Action (Review / Notice).
 - *Right 35%*: Top Non-Compliant Offender Categories (FMCG Snacks, Cosmetics, Imported Electronics, Infant Food) & Geographic Violation Heatmap.
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4.2 Screen 2: Field Inspector Mobile/Tablet Capture HUD (/capture)

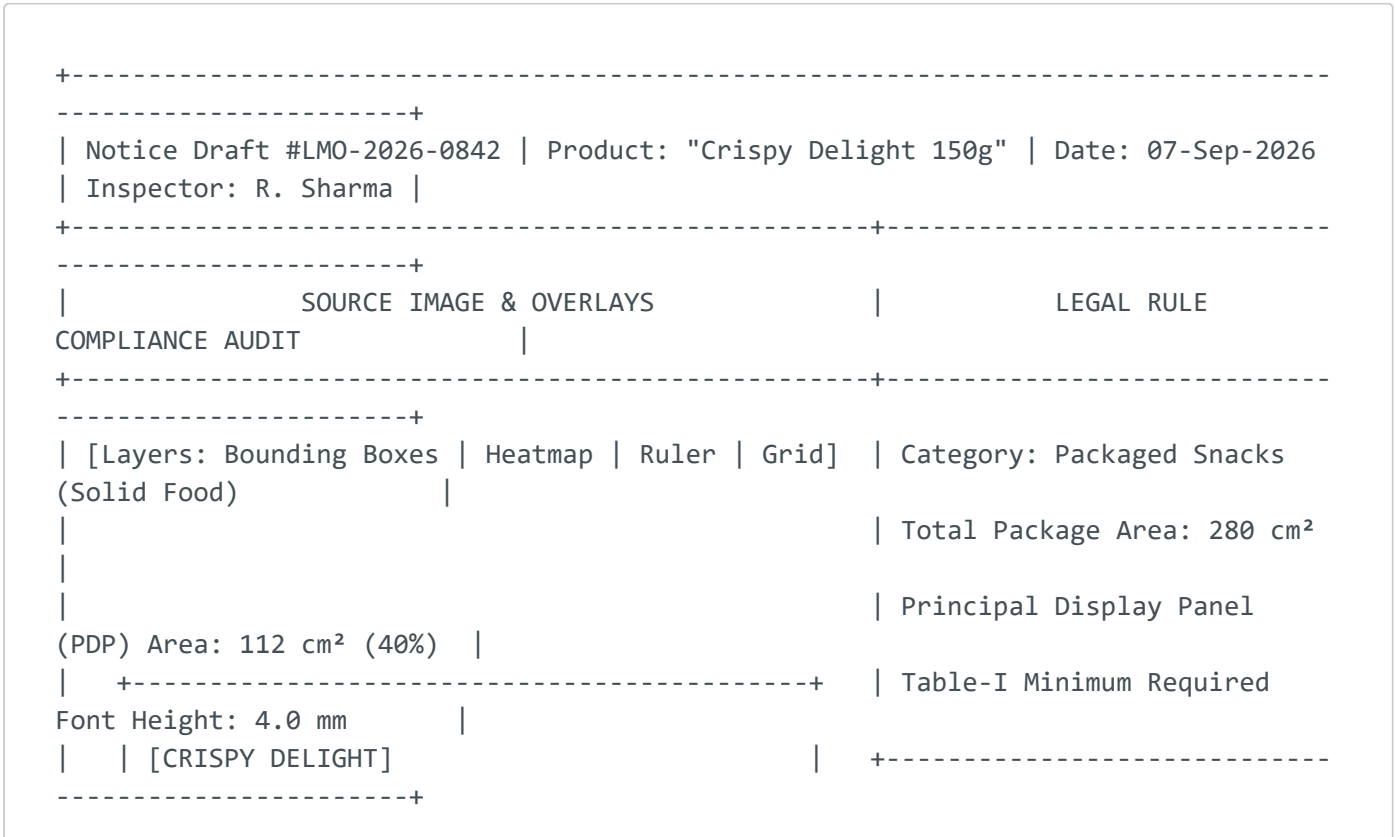
- **Primary Goal**: Guide physical label capture to ensure mathematical solvability before transmission to inference pipeline.
- **Orientation & Viewport**: Portrait & Landscape adaptive. Full-screen camera stream viewfinder.
- **Live Overlay HUD Elements**:
 1. *ArUco / Reference Ruler Anchor Box*: Semi-transparent cyan rectangle indicating where the standard reference card/coin/ruler must be placed.
 2. *Package Silhouette Guide*: Dynamic dashed outline indicating Principal Display Panel (PDP) bounding framing.
 3. *Real-Time Angle & Tilt Gyro Sensor*:
 - Pitch/Roll Indicator: Turns GREEN when tilt angle $\leq 5^\circ$ perpendicular to label.
 - Warning Toast: "Tilt too high (14°). Straighten camera to avoid projective distortion."
 4. *Glare & Specular Reflection Warning*:
 - Real-time histogram detector highlights washed-out text regions with magenta flashing border: "Excessive glare on MRP text. Reposition lighting."
 5. *Resolution / Sharpness Gate*:
 - Real-time Laplacian variance meter: "Image too blurry ($\sigma^2 < 120$). Hold steady."

6. *Trigger Button*: Large circular haptic-enabled shutter with multi-shot burst mode (Front PDP, Side Net Qty Panel, Back MRP/Mfg Panel).



4.3 Screen 3: Verification & Adjudication Canvas (/inspect/:id)

- **Primary Goal:** Dual-pane inspector verification environment. Ensures that AI never issues legal notices autonomously without an explicit human officer sign-off.
- **Layout:** Split-screen layout (Left 50% Image Loupe, Right 50% Rule Adjudication Ledger).



		MANDATORY DECLARATIONS AUDIT
(Rule 6)		
Net Weight: 150 g		

Manufacturer		[PASS] 1. Name & Address of
[FAIL: Font Height: 2.1mm < Req: 4.0mm]		Extracted: "Haldiram
Snacks Pvt Ltd..."		
		Confidence: 98.2%
Verified: [x]		
MRP Rs. 35.00		
(incl. of all taxes)		[FAIL] 2. Net Quantity
Declaration (Rule 6(1)(h))		
Unit Sale Price: Rs. 0.23 / g		Extracted: "Net
Weight: 150 g"		
		Measured Height: 2.12
mm (Req: >= 4.0 mm)		
Mfg Date: 08/2026 Use By: 02/2027		Violation: Rule 6(1)
(h) read with Table-I		
		Discrepancy: -1.88 mm
(-47.0%)		
+-----+		
		[PASS] 3. Maximum Retail
Price (Rule 6(1)(e))		
Zoom: [100%] [+] [-] [Reset] [Loupe Tool]		Extracted: "Rs. 35.00
(incl. of all taxes)"		
Measurement Mode: [Calibrated ArUco (12.4 px/mm)]		
Pixel Loupe: 142 px = 2.12 mm at standard scale		[PASS] 4. Unit Sale Price
(USP) (Rule 6(1)(k))		
		Extracted: "Rs. 0.23
/ g" (Correctly calc)		
		[PASS] 5. Consumer Care
Contacts (Rule 6(1)(n))		
		Email & Phone present
& valid format		
		[PASS] 6. Country of Origin
(Rule 6(1)(p))		
		Extracted: "Made in
India"		
+-----+		
-----+		
ACTIONS & ADJUDICATION:		
Officer Adjudication: () Approve All Compliant (*) Confirm Violation & Issue		
Legal Notice		
Officer Remarks: "Confirmed non-compliance on Net Quantity font size. Physical		
measurement agrees."		
[Generate Form-1 Legal Notice] [Request Rescan] [Reject AI Finding] [Save		
Evidence Bundle]		
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4.4 Screen 4: E-Commerce Listing Compliance Inspector (/ecommerce)

- **Primary Goal:** Ingest e-commerce product pages, analyze text attributes and product gallery images, and check Rule 6(10) and Rule 6(10A) compliance.
- **Input Methods:**
 - *Tab A: Single Product URL / DOM Ingestion:* Pastes Amazon/Flipkart/Blinkit/Zepto product URL or uploads saved HTML DOM snapshot.
 - *Tab B: Listing Gallery Image Ingestion:* Uploads the primary packaging images displayed to the online consumer.
 - *Tab C: Batch CSV / Catalog Ingestion:* Uploads listing URLs for scheduled headless ingestion.
- **E-Commerce Results View:**
 - *Declaration Completeness Checklist:*
 1. Name of the Manufacturer / Packer / Importer: Identified in DOM text? Identified on label? Match?
 2. Country of Origin: Clearly declared in tabular specifications? (Mandatory under Rule 6(10)).
 3. Net Quantity: Declared in standard units?
 4. Best Before / Expiry Date: Declared? (For perishable/package food).
 5. MRP & Unit Sale Price: Displayed clearly alongside purchase button?
 - *Rule 6(10A) 2026 Filter Compliance Audit:*
 - Checks whether the host platform exposes Country of Origin as a structured, searchable, and sortable search facet.
 - *Cross-Reference Reconciliation Widget:*
 - Compares text declared on the e-commerce webpage against the OCR text read from the uploaded gallery packaging image.
 - Flags *Discrepancy Violations* (e.g. Webpage states "Net Qty: 200g", but package image states "Net Qty: 150g").

4.5 Screen 5: Legal Notice & Report Generator (/notice/:id)

- **Primary Goal:** Automated generation of statutory legal notices under Section 36(1) of the Legal Metrology Act, 2009, complete with Section 63 BSA 2023 Digital Certificates.
- **Interactive Document Preview:**

- Standard DoCA Legal Metrology Notice Template (Form-1 / Form-2).
 - Pre-filled fields:
 - Addressee: Packer/Manufacturer name and registered corporate address.
 - Notice Reference Number: Automatically generated hierarchical format `LMO/{STATE}/{DISTRICT}/{YEAR}/{AUTO_INC}`.
 - Facts of the Case: Product seized/scanned, inspection location, date, timestamp.
 - Specific Statutory Contraventions: Citing specific Legal Metrology (Packaged Commodities) Rules, 2011 violated (e.g., Rule 6(1)(h), Rule 12, Rule 18(1)).
 - Photographic Evidence Exhibits: Crop of the non-compliant packaging panel, pixel loupe measurement diagram, OCR transcript, calibration scale metadata.
 - Mandatory Reply Period: 15 days from receipt of notice.
 - **Digital Evidence Appendices:**
 - *Annexure-A: Technical Inspection Report:* Full telemetry, model version, calibration scale, ArUco detection confidence.
 - *Annexure-B: Section 63 BSA 2023 Certificate:*
 - Device identifier, IP address, GPS coordinate coordinates (± 4 m accuracy).
 - SHA-256 hash of raw input image.
 - SHA-256 hash of generated evidence PDF bundle.
 - Digital signature of the inspecting officer.
 - **Export Formats:**
 - Signed PDF/A (Archival grade, tamper-evident).
 - JSON Evidence Bag (ZIP bundle containing raw image, cropped bounding boxes, metadata, OCR JSON, and Section 63 certificate).
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4.6 Screen 6: Batch Audit & Model Diagnostics Console (`/admin`)

- **Primary Goal:** Engineering and supervisory administration of inference pipelines, model accuracy metrics, and benchmark telemetry.
- **Components:**
 - *Inference Latency Breakdown:* Bar chart showing milliseconds spent in Preprocessing -> Panel Detection -> Text Detection (DBNet++) -> OCR (PP-OCrv4) -> Rule Engine.

- *Confidence Threshold Slider Controls*: Real-time tuning of detection thresholds with ROC curve preview.
 - *False Positive / Human Override Feedback Loop*: Lists all inspections where the human officer overrode the AI finding. Allows exporting misclassified crops directly into the fine-tuning dataset pipeline.
 - *Model Registry Status*: Active weights, ONNX INT8 quantization status, memory footprint.
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5. Accessibility, Ergonomics & Offline Sync

1. WCAG 2.1 AA Compliance:

- Contrast ratio $\geq 4.5 : 1$ for all standard text; $\geq 3.0 : 1$ for large text and bounding indicators.
- Screen reader attributes (`aria-label`, `aria-live="polite"` for real-time capture guidance).
- Full keyboard navigation support on adjudication canvas (Arrow keys to navigate bounding boxes, Space to zoom, Enter to approve, Tab to remarks).

2. Field Offline Resiliency:

- Progressive Web App (PWA) / Local SQLite Cache.
- If an inspector enters a basement godown with zero cellular connectivity, all capture guidance, local ArUco validation, and encrypted storage execute locally.
- Background sync queue uploads inspections automatically upon network reconnection.

3. Touch Ergonomics (Field Mode):

- Shutter button positioned in bottom center thumb zone.
- Critical status toasts positioned in top 15% of viewport to prevent obscuring viewfinder framing.